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Mixture factor analysis with distance metric constraint for dimensionality reduction

Jian Chen, Leiyao Liao, Wei Zhang, Lan Du*

National Lab of Radar Signal Processing, Xidian University, Xi'an 710071, China

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A B S T R A C T

Dimensionality reduction (DR) is a key preprocessing stage in high-dimensional data classification. Traditional linear DR algorithms, e.g., Linear Discriminant Analysis, transform the original data into a low-dimensional subspace with a linear transformation matrix. However, these methods cannot handle complex nonlinearly separable data. Although some nonlinear DR methods, e.g., Locally Linear Embedding, are proposed to solve this problem, most of them are unsupervised, which only focus on the data structure hidden in the original high-dimensional space, rather than maximizing the inter-class separability of the transformed data, thus reducing the classification accuracy. To tackle this challenge, a novel supervised nonlinear DR algorithm, distance metric restricted mixture factor analysis (DMR-MFA), is proposed for high-dimensional data classification. In DMR-MFA, the original data is divided into several clusters, and the generation of original data in each cluster is described via a factor analysis model. Meanwhile, the distance metric constraint (DMC) is used for maximizing the separability of transformed low-dimensional data from different classes. Moreover, the optimal model parameters are learned via the joint optimization of log-likelihood function and DMC loss function, which makes the DMR-MFA possible to obtain the more separable low-dimensional embeddings while accurately describing the original data. Experimental results on synthetic data, benchmark datasets and high-resolution range profile data demonstrate that our method can handle nonlinearly separable data and improves the classification accuracy of data with high dimensionality.

1. Introduction

With the great increasing of the number of sensors or indicators that are used to record the behavior of the objects, real-world data, such as speech signals, digital photographs, or fMRI scans, usually has a high dimensionality. How to accurately and efficiently classify high-dimensional data has become a research hotspot in many fields, such as the face recognition [1,2], image classification [3–5] and radar target recognition [6,7]. In general, the information in data with high dimensionality are redundant, which not only increases the computation burden, but also seriously affects the classification accuracy. Therefore, dimensionality reduction (DR) is beneficial for the classification of high-dimensional data, since it mitigates the curse of dimensionality and other undesired properties of high-dimensional space.

DR is the transformation of high-dimensional data into a meaningful low-dimensional representation [8]. Traditionally, DR is performed by using linear techniques, such as Principal Component Analysis (PCA) [9], Linear Discriminant Analysis (LDA) [10,11], Median-Mean Line based Discriminant Analysis (MMLDA) [12], Information Discriminant Analysis (IDA) [13], and Large Margin Nearest Neighbor (LMNN) [14]. The PCA is a commonly used linear DR method, which maximizes the variance of the transformed data. However, the between-class separability of the transformed data is ignored in PCA, which will seriously affect the classification accuracy. To solve this problem, some supervised linear DR methods are proposed. A well-known supervised linear DR method is LDA, which maximizes the ratio of between-class distance to within-class distance of the transformed data. A drawback of LDA is that the dimensionality of the transformed data must less than the number of data classes, thus restricting the application of LDA. To overcome this disadvantage, the MMLDA, an improved LDA, is proposed in [12]. By introducing the median-mean line, the MMLDA defines a new type of between-class scattering matrix, which is non-singular for any dimensionality of the transformed data. Therefore, there is no restriction for the transformed space size any more in MMLDA. Different from LDA and MMLDA, the LMNN is another category of supervised linear DR method based

* Corresponding author at: 2 Taibai Rd., Xi'an, Shaanxi, 710071, China.
E-mail address: dulan@mail.xidian.edu.cn (L. Du).

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on Mahalanobis distance metric learning [15]. The goal of LMNN is to seek a transformed low-dimensional subspace, in which the k-nearest neighbors of the transformed data belong to the same class, while the transformed data from different classes are separated by a large margin. Besides LDA, MMLDA and LMNN, some supervised linear DR methods based on the information-theoretic criteria can also obtain the better performance than PCA. The idea of these methods is to maximize the mutual information (MI) between the transformed data and their class labels, since the larger the MI is, the tighter the bound of Bayes error rate will be [16]. However, the MI between the transformed data and their class labels is difficult to calculate directly for non-Gaussian distributions. Therefore, the IDA approximates the entropy of the Gaussian mixture model (GMM) with the entropy of a global Gaussian distribution, and provides an approximate expression to the Shannon MI. The mutual information maximization (MIM) in [17] gives an analytic and explicit expression for the gradient of the Shannon MI with respect to the transformation matrix for any distribution, and thus, the Shannon MI can be maximized directly without any approximation. Based on the IDA, an enhancing IDA (EIDA) is proposed in [18]. EIDA describes the statistical structure of the transformed low-dimensional subspace via a linear statistical model, and utilizes the MI based information-theoretic criterion to further maximize the inter-class separability.

Although above linear DR methods are widely used in high-dimensional data classification, they cannot adequately handle complex nonlinearly separable data. To solve this problem, several nonlinear DR techniques have been proposed in the past years. Typical examples include Isometric Feature Mapping (Isomap) [19], Locally Linear Embedding (LLE) [20], kernelized PCA (KPCA) [21] and Mixture Factor Analysis (MFA) [22]. The goal of Isomap is that the geodesic distance of each pair from the original data is equal to that from the transformed low-dimensional data. In Isomap, the geodesic distances between the original data can be computed using Dijkstra's or Floyd's shortest-path algorithm [23,24], thereby forming a pairwise geodesic distance matrix. The transformed low-dimensional data are computed by applying classical scaling on the resulting pairwise geodesic distance matrix. In contrast to Isomap, the LLE attempts to preserve solely local properties of the original data into the transformed low-dimensional space. In detail, the LLE thinks that there is a linear correlation among the original data in a local region. Thus for each sample $\mathbf{x}_i$ ($i = 1, \cdots, N$ and N denotes the data size), it can be written as $\mathbf{x}_i = \sum_j w_{ij}\mathbf{x}_{ij}$, where $\mathbf{x}_{ij}$ represents the jth nearest neighbor of $\mathbf{x}_i$, $j = 1, \cdots, J$ with J denoting the number of nearest neighbors, $\mathbf{w}_i$=$[w_{i1}, \cdots, w_{iJ}]$ represents the weight to reconstruct $\mathbf{x}_i$ and can be solved via the minimization of reconstruction error, i.e., $\min \sum_i \|\mathbf{x}_i - \sum_j w_{ij}\mathbf{x}_{ij}\|_2^2$. Meanwhile, the LLE assumes that the neighbor structure of data in the transformed low-dimensional space is consistent with that in the original space. In other words, the weight $\mathbf{w}_i$ can also be used for the nearest neighbor-based reconstruction of the low-dimensional representation of $\mathbf{x}_i$, denoted as $\mathbf{y}_i$. Consequently, the data DR in the LLE can be completed by minimizing the expression of $\sum_i \|\mathbf{y}_i - \sum_j w_{ij}\mathbf{y}_{ij}\|_2^2$, with $\mathbf{y}_{ij}$ denoting the low-dimensional projection of the nearest neighbor $\mathbf{x}_{ij}$. The KPCA is the reformulation of traditional linear PCA in a kernel space that is constructed using a kernel function. The KPCA computes the principal eigenvectors of the kernel matrix rather than those of the covariance matrix in PCA. The application of PCA in the kernel space provides KPCA the property of constructing nonlinear mapping. The MFA is another type of nonlinear DR method. In MFA, the global nonlinearly separable data is divided into several clusters, and the transformed low-dimensional data in each cluster is given with a linear DR method, i.e., factor analysis (FA) [25]. With the combination of several linear models, the nonlinear mapping is achieved in MFA. Although these nonlinear DR methods can handle the complex nonlinearly separable data, they are unsupervised and only focus on the data structure hidden in the original high-dimensional space, rather than maximizing the between-class separability of the transformed low-dimensional data, which reduces the accuracy of classification.

In order to tackle such challenge, a supervised nonlinear DR method, referred to as the distance metric restricted mixture factor analysis (DMR-MFA), is proposed for high-dimensional data classification in this paper. The DMR-MFA is an improved version of MFA by imposing the distance metric constraint (DMC) on transformed low-dimensional space. Same as the MFA, the DMR-MFA divides the original data into several clusters, and describes the statistic structure of the transformed space in each cluster with a FA model. Simultaneously, the DMC, i.e., the Euclidean distance of the transformed low-dimensional data from the same class is minimal and that from different classes is maximal, is utilized to further maximize the inter-class separability of the transformed data. Moreover, the log-likelihood function of MFA and DMC loss function are jointly optimized to seek the optimal model parameters.

The main contributions of our method are summarized as follows.

- In the combination of the merits of MFA and distance metric learning, the proposed DMR-MFA model has the potential to learn low-dimensional representations with the stronger separability from nonlinearly separable data, compared with the existing DR methods transforming all data via a simple linear projection or only focusing on the structure description of representations rather than their inter-class differences in the low-dimensional space.
- More importantly, the proposal of our DMR-MFA provides a set of line of thought that one can better deal with the problem of low-dimensional feature extraction by adopting the combination of distance metric learning and unsupervised nonlinear model.
- We comprehensively exploit the expectation maximization and gradient ascent algorithms to learn the model parameters, which makes the joint optimization of log-likelihood from MFA and distance metric constraint loss possible.

In Table 1, we list all notations in our DMR-MFA model to assist readers clearly understand the matrix dimensions and definitions of notations.

Our paper is organized as follows. Section 2 gives a brief overview of MFA, distance metric learning, and Bayesian classifier. The proposed method is introduced in Section 3. Section 4 evaluates the performance of our method on synthetic data, benchmark datasets, and high-resolution range profile (HRRP) data. Section 5 summarizes the paper by presenting some concluding remarks.

2. Background

2.1. MFA model

FA is a method for modeling correlations in multidimensional data. The model assumes that each d-dimensional data $\mathbf{x}_i$ is generated by first linearly transforming m-dimensional (usually $m < d$) vector $\mathbf{y}_i$ and then adding a d-dimensional Gaussian noise variable $\boldsymbol{\varepsilon}_i$. Mathematically, the FA model can be expressed as

$$\mathbf{x}_i = \mathbf{A}\mathbf{y}_i + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_i \tag{1}$$

where $\boldsymbol{\mu} \in \mathbb{R}^d$ is the mean vector of all observed samples $\{\mathbf{x}_i\}_{i=1}^N$, and $\mathbf{A} \in \mathbb{R}^{d\times m}$ is the factor loading matrix; $\mathbf{y}_i \in \mathbb{R}^m$ is the latent

Table 1
List of notations in the DMR-MFA model.

Notations	Meanings	Notations	Meanings
N	The number of observed samples	$\mathbf{y}_{uk} \in \mathbb{R}^m$	The low-dimensional latent feature of $\mathbf{x}_{uk}$
K	The number of clusters	$\mathbf{A}_k \in \mathbb{R}^{d\times d}$	Factor loading matrix for the cluster k
n_k	The sample number in the kth cluster with k=1, $\cdots$, K and $\sum_{k=1}^{K} n_k = N$	$\boldsymbol{\mu}_k \in \mathbb{R}^d$	Mean vector of observed samples in the kth cluster
d	The dimension of observed sample	$\boldsymbol{\varepsilon}_k \in \mathbb{R}^d$	Noise variable for the cluster k with the mean of $\mathbf{0}$ and covariance matrix of $\boldsymbol{\Psi}_k$
m	The dimension of latent feature	$\mathbf{C}_k \in \mathbb{R}^{d\times d}$	Covariance matrix of cluster k and $C_k = (\mathbf{A}_k\mathbf{A}_k^T + \boldsymbol{\Psi}_k)$
$\mathbf{x}_i \in \mathbb{R}^d$	The ith observed sample with $i = 1, \cdots, N$	$l_{uv} \in \{-1, 1\}$	$l_{uv} = 1$ if two samples $\mathbf{x}_{uk}$ and $\mathbf{x}_{vk}$ in cluster k belong to the same class; otherwise, $l_{uv} = -1$
$\mathbf{y}_i \in \mathbb{R}^m$	The low-dimensional latent feature of $\mathbf{x}_i$	τ	The threshold used in the distance metric constraint
$z_i \in \{1, \cdots, K\}$	Cluster indicator of $\mathbf{x}_i$	$J_{\mathrm{DMC}}(\theta)$	The distance metric constraint function with θ denoting the parameters in the DMR-MFA
$\gamma_{ik} \in \{0, 1\}$	$\gamma_{ik} = I(z_{ik} = k)$ with $I(\cdot)$ being the indicator function	$L_{\mathrm{MFA}}(\theta)$	Log-likelihood function of the MFA module
$\boldsymbol{\pi} = [\pi_1, \cdots, \pi_K]$	Mixing proportion vector with $\sum_{k=1}^{K} \pi_k = 1$	$\mathbf{L}_{\mathrm{DMR-MFA}}(\theta)$	The objective function of the DMR-MFA model
$\mathbf{x}_{uk} \in \mathbb{R}^d$	The uth sample in the kth cluster and $u = 1, \cdots, n_k$	λ	Weighting factor between the $J_{\mathrm{DMC}}(\theta)$ and $L_{\mathrm{MFA}}(\theta)$

variable and assumed to be Gaussian distributed as $N(\mathbf{0}, \mathbf{I})$ with its mean $\mathbf{0}$ being a zero vector and covariance matrix $\mathbf{I}$ being an identity matrix; the noise variable $\boldsymbol{\varepsilon}_i \in \mathbb{R}^d \sim N(\mathbf{0}, \boldsymbol{\Psi})$ and $\boldsymbol{\Psi}$ is a diagonal matrix with the diagonal elements being different. According to the property of Gaussian distribution, we have $p(\mathbf{x}_i) = N(\mathbf{x}_i|\boldsymbol{\mu}, \mathbf{A}\mathbf{A}^T + \boldsymbol{\Psi})$. Given a dataset $X = \{\mathbf{x}_i\}_{i=1}^N$, the log-likelihood of the FA model can thus be derived as

$$L(\theta) = \sum_{i=1}^{N} \ln p(\mathbf{x}_i) = -\frac{Nd}{2}\ln(2\pi) - \frac{N}{2}\ln\left|\left(\mathbf{A}\mathbf{A}^T + \boldsymbol{\Psi}\right)\right| - \frac{1}{2}tr\left[\left(\mathbf{A}\mathbf{A}^T + \boldsymbol{\Psi}\right)^{-1}\left(\sum_{i=1}^{N}(\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T\right)\right], \tag{2}$$

where $\theta = \{\mathbf{A}, \boldsymbol{\mu}, \boldsymbol{\Psi}\}$, N represents the number of samples, and $tr(\cdot)$ is the trace of a matrix. The optimal parameters $\mathbf{A}^*$ and $\boldsymbol{\Psi}^*$ can be obtained by maximizing the log-likelihood function, i.e., $\{\mathbf{A}^*, \boldsymbol{\Psi}^*\} = \arg\max_{\mathbf{A}, \boldsymbol{\Psi}} L(\theta)$.

Considering a FA for each component of Gaussian Mixture Model (GMM), clustering and local DR can be addressed simultaneously by MFA, which models the density of $\mathbf{x}_i$ as a weighted average of FA densities, i.e.,

$$\begin{aligned} p(\mathbf{x}_i) &= \int p(z_i = k|\boldsymbol{\pi})p(\mathbf{x}_i|\mathbf{y}_i, z_i)p(\mathbf{y}_i|z_i)d\mathbf{y}_i \\ &= \int \pi_k N(\mathbf{x}_i; \mathbf{A}_{z_i}\mathbf{y}_i + \boldsymbol{\mu}_{z_i}, \boldsymbol{\Psi}_{z_i})\, N(\mathbf{y}_i; 0, \mathbf{I})d\mathbf{y}_i \\ &= \pi_k N\left(\mathbf{x}_i; \boldsymbol{\mu}_{z_i}, \mathbf{A}_{z_i}\mathbf{A}_{z_i}^T + \boldsymbol{\Psi}_{z_i}\right), \end{aligned}$$

where $z_i \in \{1, \cdots, K\}$ is a discrete indicator variable with K being the number of clusters; $p(z_i = k|\boldsymbol{\pi}) = \pi_k$ and $\boldsymbol{\pi} = [\pi_1, \ldots, \pi_K]$ is the vector of mixing proportions with $\sum_{k=1}^{K} \pi_k = 1$; $\mathbf{A}_{z_i}$, $\boldsymbol{\mu}_{z_i}$ and $\boldsymbol{\Psi}_{z_i}$ are the factor loading matrix, mean vector and diagonal covariance matrix of Gaussian noise variable in the z_ith FA model, respectively. For a clearer illustration, we give the graphical model of MFA in Fig. 1.

Assuming the latent variable $\gamma_{ik} = I(z_i = k)$ with $I(\cdot)$ being an indicator function, the log-likelihood function of MFA is

$$\begin{aligned} L_{\mathrm{MFA}}(\theta) &= \sum_{i=1}^{N} \ln p(\mathbf{x}_i, \gamma_{i1}, \cdots, \gamma_{iK}) = \sum_{i=1}^{N}\sum_{k=1}^{K} \ln\left[p(\mathbf{x}_i)\right]^{\gamma_{ik}} \\ &= \sum_{k=1}^{K}\left\{n_k \ln \pi_k + \sum_{i=1}^{N} \gamma_{ik} \ln\left[N\left(\mathbf{x}_i; \boldsymbol{\mu}_k, \mathbf{A}_k\mathbf{A}_k^T + \boldsymbol{\Psi}_k\right)\right]\right\}, \end{aligned} \tag{4}$$

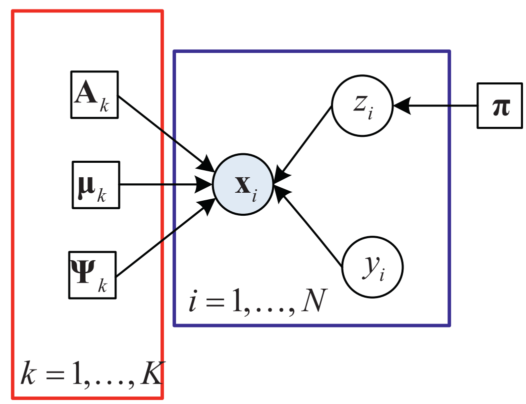


Fig. 1. The graphical model of MFA. The left part framed with the red box consists of the local variables related to cluster k, the right part framed with the blue box is composed of the local variables related to observed sample $\mathbf{x}_i$, and $\boldsymbol{\pi}$ is a global variable shared among all data. Expressions $k = 1, \cdots, K$ and $i = 1, \cdots, N$ at the edge of a box indicate that there is a stack of such boxes for the indexes k and i, where K and N denote the number of clusters and samples, respectively. The detailed generative process of $\mathbf{x}_i$ via the MFA is expressed in (1) and (3).

where $n_k = \sum_{i=1}^{N} \gamma_{ik}$ and $\sum_{k=1}^{K} n_k = N$. The parameters of MFA model $\theta = \{(\mathbf{A}_k, \boldsymbol{\mu}_k, \boldsymbol{\Psi}_k)_{k=1}^K, \boldsymbol{\pi}\}$ can be optimized via the Expectation Maximization (EM) algorithm [26].

2.2. Distance metric learning

Distance metric learning defines a distance metric to measure the similarity of pair data. A well-known distance metric learning algorithm is Mahalanobis Distance metric learning (MDML), which defines the distance of the data $\mathbf{x}_i$ and $\mathbf{x}_j$ as

$$Dis\left(\mathbf{x}_i, \mathbf{x}_j\right) = \sqrt{\left(\mathbf{x}_i - \mathbf{x}_j\right)^T \mathbf{M}\left(\mathbf{x}_i - \mathbf{x}_j\right)}, \tag{5}$$

where $\mathbf{M}$ is a positive semi-definite metric matrix. The goal of MDML is to minimize (maximize) the Mahalanobis distance of the data with the same (different) class labels. Since $\mathbf{M}$ is a positive demi-definite matrix, it can be rewritten as

$$\mathbf{M} = \mathbf{P}^T\mathbf{P}. \tag{6}$$

Substituting (6) into (5), the Mahalanobis Distance metric can be represented as

$$Dis\left(\mathbf{x}_i, \mathbf{x}_j\right) = \sqrt{\left(\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j\right)^T\left(\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j\right)}. \tag{7}$$

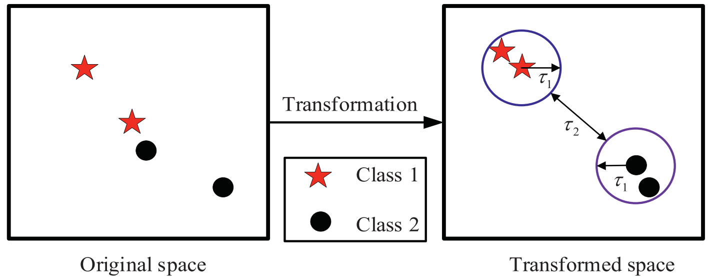


Fig. 2. The explanation of Mahalanobis Distance metric learning (MDML). Here the red stars and black solid circles are the sample points from two different classes, where the distance between two samples with different class labels may smaller than those of samples with the same class label in the original space. Via the MDML, the intra-class distance is smaller than a threshold τ_1 and the inter-class distance is larger than a threshold τ_2 ($\tau_2 > \tau_1$) in the transformed space.

We see from (7) that the Mahalanobis Distance of the input space is equivalent to the Euclidean distance of the transformed space. This equivalence suggests that the estimation of $\mathbf{M}$ can be turned into the estimation of $\mathbf{P}$. In order to enlarge the similarity of transformed data with the same class label and reduce the similarity of transformed data with different class labels, the distance metric constraint (DMC), i.e., the distance of transformed samples from the same class is smaller than a threshold τ_1 and that from different classes is larger than a threshold τ_2 ($\tau_2 > \tau_1$), is used to ensure the separability of features. Fig. 2 gives an explanation about this idea.

Based on the above analysis, the DMC can be represented as follows:

$$\begin{aligned} dis_{\mathbf{P}}^2(\mathbf{x}_i, \mathbf{x}_j) &= (\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j)^T(\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j) \le \tau_1 \text{ if } l_{ij} = 1, \\ dis_{\mathbf{P}}^2(\mathbf{x}_i, \mathbf{x}_j) &= (\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j)^T(\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j) \ge \tau_2 \text{ if } l_{ij} = -1, \end{aligned} \tag{8}$$

where l_{ij}=1 represents that samples $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to same class; otherwise, $l_{ij} = -1$. Suppose τ_1=$\tau - 1$ and τ_2=τ+1, (8) can be rewritten as $l_{ij}(\tau - \|\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j\|_2^2) \ge 1$. Therefore, the loss of DMC can be obtained as follows

$$\mathcal{J}_{\mathrm{DMC}}(\mathbf{P}) = \min \sum_{i,j} \max\left(0, 1 - l_{ij}\left(\tau - \left\|\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j\right\|_2^2\right)\right). \tag{9}$$

Since the function $[z]_+ = \max(z, 0)$ is not a smooth function, the generalized logistic loss function $g(z) = \beta^{-1} \ln[1 + \exp(\beta z)]$ is used as a smoothed approximation of $[z]_+$. Therefore, the objective function of DMC can further be expressed as

$$\mathcal{J}_{\mathrm{DMC}}(\mathbf{P}) = \min_{\mathbf{P}} \sum_{i,j} g\left(1 - l_{ij}\left(\tau - \left\|\mathbf{P}\mathbf{x}_i - \mathbf{P}\mathbf{x}_j\right\|^2\right)\right). \tag{10}$$

Finally, the transformed matrix $\mathbf{P}$ can be optimized via the gradient descent on (10).

2.3. Bayesian classifier

Bayesian classifier [27] is a widely used classifier to evaluate the performance of supervised DR. Suppose $c \in \{1, \ldots, C\}$ denotes a discrete-valued class variable with C representing the number of classes, $\mathbf{x}$ denotes an random variable, the prior distribution of class variable is set to $p(c) \triangleq w_c$ for $\forall c = 1, 2, \ldots, C$, and the class-conditional probability density function (PDF) is set as Gaussian distribution with the class mean vector $\boldsymbol{\mu}_c$ and class covariance matrix $\boldsymbol{\Sigma}_c$, i.e., $p(\mathbf{x}|c) \sim N(\mathbf{x}; \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$. According to Bayesian theorem, the posterior distribution of class variable can be written as $p(c|\mathbf{x}) = p(\mathbf{x}|c)w_c/p(\mathbf{x})$ with $p(\mathbf{x}) = \sum_{c=1}^{C} p(\mathbf{x}|c)w_c$.

The essence of Bayesian classifier is to classify a test sample $\mathbf{x}^*$ into the class c^* with the maximum class posterior probability, i.e., $c^* = \arg \max_{c=1,2,\ldots,C} p(c|\mathbf{x}^*)$. Since $p(\mathbf{x}^*)$ is a constant with respect to c, the above expression is equivalent to

$$c^* \propto \arg \max_{c=1,2,\ldots,C} \ln\left[p(\mathbf{x}^*|c)w_c\right], \tag{11}$$

where $\ln[p(\mathbf{x}^*|c)w_c]$=$[-\frac{d}{2}\ln(2\pi) - \frac{1}{2}\ln|\boldsymbol{\Sigma}_c| - \frac{1}{2}(\mathbf{x}^* - \boldsymbol{\mu}_c)^T\boldsymbol{\Sigma}_c^{-1}(\mathbf{x}^* - \boldsymbol{\mu}_c) + \ln(w_c)]$. For a clearer illustration, we give the flowchart of classification method based on Bayesian classifier in Fig. 3, from which we note that the integrated scheme includes training stage and testing stage. Then we introduce the two stages in detail.

- Training stage

In this stage, the training data firstly need to be preprocessed, such as normalization operation. Thereafter, a model is constructed for all observations (such as the MFA) and the parameter estimation is subsequently performed via an appropriate solution algorithm. Once the model learning completes, the class mean $\boldsymbol{\mu}_c$ and class covariance matrix $\boldsymbol{\Sigma}_c$ ($c = 1, 2, \ldots, C$) can be estimated based on the learned model parameters and should be stored as templates for the Bayesian classifier construction in the testing phase.

- Test stage

In the test stage, after data preprocessing, the test sample is fed into the constructed Bayesian classifier, with which the class posterior probabilities of the test sample given all classes can be calculated according to (11). Ultimately, this unknown sample is assigned to the class with the maximum class posterior probability.

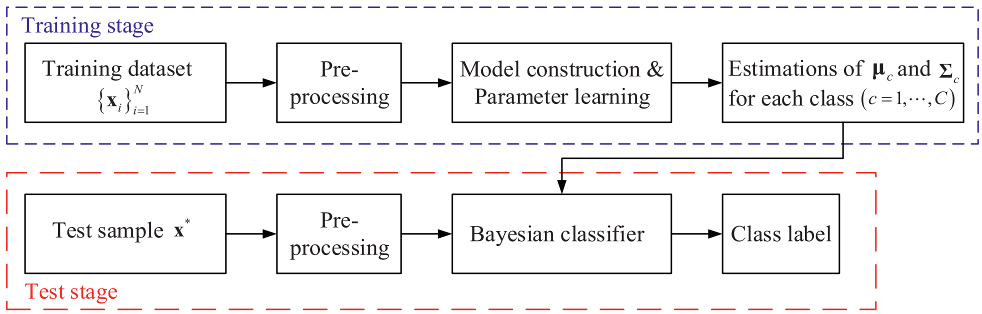


Fig. 3. The flowchart of classification based on Bayesian classifier. The training stage framed by the blue dotted line mainly realizes the estimations of class mean $\boldsymbol{\mu}_c$ and class covariance matrix $\boldsymbol{\Sigma}_c$ for each class c ($c = 1, \cdots, C$) via the training data. The test stage framed by the red dotted line makes the prediction for the class membership of a test sample via the Bayesian classifier constructed by the estimated parameters in the training stage.

3. Distance metric restricted mixture factor analysis model

3.1. Objective function of the DMR-MFA

Traditional supervised linear DR methods cannot handle the complex nonlinearly separable data, such as the LDA and LMNN. To deal with this issue, some nonlinear DR algorithms, e.g., LLE and MFA, are proposed in the past years. However, since these methods are unsupervised, the inter-class separability of transformed low-dimensional data is not maximized, which reduces the accuracy for high-data classification. In this section, we propose a supervised nonlinear DR technology for high-dimensional data classification. The proposed method, i.e., DMR-MFA, is an improved version of MFA by imposing the DMC on the transformed low-dimensional data. In DMR-MFA, the generation of the high-dimensional data is described by MFA. Meanwhile, the DMC, i.e., the distance of the transformed data from the same class is smaller than a threshold and that from different classes is larger than a threshold, is imposed on the low-dimensional data to maximize their inter-class separability.

As introduced in Section 2.1, the MFA is a mixture-of-expert model, which divides the original high-dimensional data into several clusters and describes the generation of the original data in each cluster via a FA model. For the uth observed sample belonging to the kth cluster $\mathbf{x}_{uk} \in \mathbb{R}^d$, the MFA represents it as

$$\mathbf{x}_{uk} = \mathbf{A}_k \mathbf{y}_{uk} + \boldsymbol{\mu}_k + \boldsymbol{\varepsilon}_{uk}, \tag{12}$$

where $\mathbf{A}_k \in \mathbb{R}^{d\times m}$, $\boldsymbol{\mu}_k \in \mathbb{R}^d$ and $\boldsymbol{\varepsilon}_{uk} \sim N(\mathbf{0}, \boldsymbol{\Psi}_k)$ are the factor loading matrix, mean vector and Gaussian noise variable for the samples in the cluster k, respectively; $\mathbf{y}_{uk} \in \mathbb{R}^m \sim N(\mathbf{0}, \mathbf{I})$ is the transformed low-dimensional latent feature of $\mathbf{x}_{uk}$, and $u = 1, \cdots, n_k$ with n_k denoting the sample number of the cluster k and $\sum_{k=1}^K n_k = N$. Then the conditional posterior of latent feature$p(\mathbf{y}_{uk}|\mathbf{x}_{uk})$ can be derived as

$$p(\mathbf{y}_{uk}|\mathbf{x}_{uk}) \sim N\left(\mathbf{y}_{uk}; \mathbf{A}_k^T \mathbf{C}_k^{-1}(\mathbf{x}_{uk} - \boldsymbol{\mu}_k), \mathbf{I} - \mathbf{A}_k^T \mathbf{C}_k^{-1} \mathbf{A}_k\right), \tag{13}$$

where $C_k = (\mathbf{A}_k \mathbf{A}_k^T + \boldsymbol{\Psi}_k)$. Given $\mathbf{x}_{uk}$, the transformed low-dimensional latent feature $\mathbf{y}_{uk}$ can be approximated with the conditional mean $\mathbf{A}_k^T \mathbf{C}_k^{-1}(\mathbf{x}_{uk} - \boldsymbol{\mu}_k)$.

In the DMR-MFA, the DMC is imposed on the latent feature in each cluster. Therefore, the latent feature $\mathbf{y}_{uk}$ should satisfy the following constraint according to (8):

$$l_{uv}\left(\tau - \|\mathbf{y}_{uk} - \mathbf{y}_{vk}\|_2^2\right) = l_{uv}\left(\tau - \left\|\mathbf{A}_k^T \mathbf{C}_k^{-1}(\mathbf{x}_{uk} - \mathbf{x}_{vk})\right\|_2^2\right) \geq 1, \tag{14}$$

where l_{uv}=1 if two different samples $\mathbf{x}_{uk}$ and $\mathbf{x}_{vk}$ in the kth cluster belong to the same class; by contrary, $l_{uv} = -1$ when $\mathbf{x}_{uk}$ and $\mathbf{x}_{vk}$ come from different classes. Referring to (10), the loss function of the DMC in our model, denoted as $\mathcal{J}_{\mathrm{DMC}}(\theta)$, can be represented as

$$\mathcal{J}_{\mathrm{DMC}}(\theta) = \min J_{\mathrm{DMC}}(\theta) = \min \sum_{u,v} g\left(1 - l_{uv}\left(\tau - \left\|\mathbf{A}_k^T \mathbf{C}_k^{-1} \mathbf{x}_{uk} - \mathbf{A}_k^T \mathbf{C}_k^{-1} \mathbf{x}_{vk}\right\|_2^2\right)\right) \tag{15}$$

where $J_{\mathrm{DMC}}(\theta) = \sum_{u,v} g(1 - l_{uv}(\tau - \|\mathbf{A}_k^T \mathbf{C}_k^{-1} \mathbf{x}_{uk} - \mathbf{A}_k^T \mathbf{C}_k^{-1} \mathbf{x}_{vk}\|_2^2))$ denotes the DMC function.

In DMR-MFA, we want to find the model parameters $\theta = \{(\mathbf{A}_k, \boldsymbol{\mu}_k, \boldsymbol{\Psi}_k)_{k=1}^K, \boldsymbol{\pi}\}$ which can not only maximize the separability among the transformed low-dimensional features from different classes but also describe the statistical structure of original data accurately. This purpose can be achieved via the optimization problem of

$$\max_{\theta} L_{\mathrm{MFA}}(\theta) - \lambda J_{\mathrm{DMC}}(\theta), \tag{16}$$

where λ denotes the weighting factor, $L_{\mathrm{MFA}}(\theta)$ is the log-likelihood function of MFA, and $J_{\mathrm{DMC}}(\theta)$ is the DMC function. As discussed in Section 2.1, we have

$$L_{\mathrm{MFA}}(\theta) = \sum_{k=1}^K \left\{ n_k \ln \pi_k + \sum_{i=1}^N \gamma_{ik} \ln\left[N\left(\mathbf{x}_i; \boldsymbol{\mu}_k, \mathbf{A}_k \mathbf{A}_k^T + \boldsymbol{\Psi}_k\right)\right]\right\}. \tag{17}$$

In combination of (15)-(17), the objective function of DMR-MFA can thus be expressed as

$$\begin{aligned}\mathbf{L}_{\mathrm{DMR-MFA}}(\theta) = \max & \underbrace{\sum_{k=1}^{K}\left\{n_k \ln \pi_k + \sum_{i=1}^{N}\gamma_{ik}\ln\left[N\left(\mathbf{x}_i;\mathbf{u}_k,\mathbf{A}_k\mathbf{A}_k^T+\boldsymbol{\Psi}_k\right)\right]\right\}}_{\text{log-likelihood function of MFA model}} \\ & -\lambda\underbrace{\sum_{k=1}^{K}\sum_{u,v}\mathrm{g}\left(1-l_{uv}\left(\tau-\left\|\mathbf{A}_k^T\mathbf{C}_k^{-1}\mathbf{x}_{vk}-\mathbf{A}_k^T\mathbf{C}_k^{-1}\mathbf{x}_{vk}\right\|_2^2\right)\right)}_{\text{loss function of DMR}}\end{aligned} \tag{18}$$

- Relationship of our method and MFA

Traditional MFA is an unsupervised generation model and can be used for nonlinear DR. The goal of MFA is to describe the statistical structure of observed data accurately by maximizing the log-likelihood function. However, the inter-class separability of the low-dimensional latent features is not considered, which limits the classification accuracy. Our method is an improved MFA, which not only describes the statistical structure of observed data accurately, but also maximizes the separability among the transformed low-dimensional features in each cluster, thus in favor of the further improvement on high-dimensional data classification.

- Relationship of our method and Mahalanobis distance metric learning (MDML)

In fact, our method can be viewed as a special class of MDML algorithm. Compared with some traditional MDML methods, our method utilizes the probabilistic decoding model, i.e.,$\mathbf{x}=\mathbf{Ay}+\boldsymbol{\mu}+\boldsymbol{\varepsilon}$ with $\mathbf{y}\sim N(\mathbf{0},\mathbf{I})$, $\boldsymbol{\varepsilon}\sim N(\mathbf{0},\boldsymbol{\Psi})$, to extract observation's low-dimensional latent feature, expressed as $\mathbf{y}_{\mathrm{DMR-MFA}}=\mathbf{A}^T\mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})$ shown in (13). However, the traditional MDML uses a algebraic encoding model, i.e., $\mathbf{y}_{\mathrm{MDML}}=\mathbf{Px}$ as shown in (7), to perform feature extraction. The $\mathbf{y}_{\mathrm{DMR-MFA}}$ considers the statistical characteristics of observed data, i.e., the mean $\boldsymbol{\mu}$ and covariance $\mathbf{C}=(\mathbf{AA}^T+\boldsymbol{\Psi})$. Compared with the point estimation of feature in the MDML, our model has the stronger generalization ability. In detail, our method can still perform well in separable feature extraction when an unseen sample comes from the same probability distribution as the known data but is greatly different from the known data. However, the performance of MDML cannot be guaranteed in this case.

3.2. Optimization procedure of DMR-MFA

In this section, we combine the EM and gradient ascent algorithms to achieve the estimation of model parameter θ=$\{(\mathbf{A}_k,\boldsymbol{\mu}_k,\boldsymbol{\Psi}_k)_{k=1}^K,\boldsymbol{\pi}\}$. The details are organized as follows.

E-step: Computing the expectation of $L_{\mathrm{MFA}}(\theta)-\lambda J_{\mathrm{DMC}}(\theta)$ ***with respect to latent variable*** $\gamma=\{\gamma_{ik}\}_{k=1,i=1}^{K\ ,\ N}$ ***given*** θ***.***

$$Q(\theta)=E_\gamma[L_{\mathrm{MFA}}(\theta)-\lambda J_{\mathrm{DMC}}(\theta)]=\sum_{k=1}^{K}\left\{\sum_{i=1}^{N}E(\gamma_{ik})\ln\pi_k+E(\gamma_{ik})\ln\left[N\left(\mathbf{x}_i;\boldsymbol{\mu}_k,\mathbf{A}_k\mathbf{A}_k^T+\boldsymbol{\Psi}_k\right)\right]\right\}-\lambda J_{\mathrm{DMC}}(\theta) \tag{19}$$

where $E_\gamma(\cdot)$ is the posterior expectation with respect to γ, $E(\cdot)$ is the posterior expectation, and $h_{ik}\triangleq E(\gamma_{ik})=\frac{\pi_k N(\mathbf{x}_i;\boldsymbol{\mu}_k,\mathbf{A}_k\mathbf{A}_k^T+\boldsymbol{\Psi}_k)}{\sum_{k=1}^{K}\pi_k N(\mathbf{x}_i;\boldsymbol{\mu}_k,\mathbf{A}_k\mathbf{A}_k^T+\boldsymbol{\Psi}_k)}$.

M-step: Maximizing $Q(\theta)$ ***to obtain the optimal estimation of*** θ ***given*** γ***.***

In this step, the traditional EM algorithm seeks an optimal estimation of θ by setting the derivate of $Q(\theta)$ on θ to zero. However, the introduction of $J_{\mathrm{DMC}}(\theta)$ makes the analytic solution of θ intractable (except $\boldsymbol{\pi}$ which has the analytic expression as $\frac{1}{\pi_k}\sum_{i=1}^{N}h_{ik}$). As an alternative way, we adopt the gradient ascent method to optimize $Q(\theta)$, where the gradients of $L_{\mathrm{MFA}}(\theta)$ and $J_{\mathrm{DMC}}(\theta)$ on θ are the key and difficult points. To implement model learning, we calculate the gradient of θ and attach the result as follows.

- Let $diag(\mathbf{S})$ represent the diagonalization operation of a matrix $\mathbf{S}$. The gradient of $L_{\mathrm{MFA}}(\theta)$ with respect to the noise variance $\boldsymbol{\Psi}_k$ ($k=1,\cdots,K$) can be expressed as

$$\frac{\partial L_{\mathrm{MFA}}(\theta)}{\partial\boldsymbol{\Psi}_k}=diag\left\{\frac{1}{N}\mathbf{C}_k^{-1}\left(\sum_{i=1}^{N}h_{ik}(\mathbf{x}_i-\boldsymbol{\mu}_k)(\mathbf{x}_i-\boldsymbol{\mu}_k)^T\right)\mathbf{C}_k^{-1}-\left(\sum_{i=1}^{N}h_{ik}\right)\mathbf{C}_k^{-1}\right\} \tag{20}$$

- The gradient of $J_{\mathrm{DMC}}(\theta)$ with respect to the noise variance $\boldsymbol{\Psi}_k$ is

$$\frac{\partial J_{\mathrm{DMC}}(\theta)}{\partial\boldsymbol{\Psi}_k}=diag\left(\frac{1}{n_k^2}\sum_{u,v}^{n_k}\frac{\exp(\beta L_{uv})}{1+\exp(\beta L_{uv})}l_{uv}\mathbf{C}_k^{-1}\mathbf{A}_k\mathbf{A}_k^T\mathbf{P}_{uv}\right) \tag{21}$$

where L_{uv}= $1-l_{uv}(\tau-\|\mathbf{A}_k^T\mathbf{C}_k^{-1}\mathbf{x}_{uk}-\mathbf{A}_k^T\mathbf{C}_k^{-1}\mathbf{x}_{vk}\|_2^2)$, and $\mathbf{P}_{uv}$=$\mathbf{C}_k^{-1}(\mathbf{x}_{uk}-\mathbf{x}_{vk})(\mathbf{x}_{uk}-\mathbf{x}_{vk})^T\mathbf{C}_k^{-1}$.

- The gradient of log-likelihood function $L_{\mathrm{MFA}}(\theta)$ with respect to the transformation matrix $\mathbf{A}_k$ can be expressed as

$$\frac{\partial L_{\mathrm{MFA}}(\theta)}{\partial\mathbf{A}_k}=\frac{1}{N}\left\{\mathbf{C}_k^{-1}\left(\sum_{i=1}^{N}h_{ik}(\mathbf{x}_i-\boldsymbol{\mu}_k)(\mathbf{x}_i-\boldsymbol{\mu}_k)^T\right)\mathbf{C}_k^{-1}\mathbf{A}_k-\left(\sum_{i=1}^{N}h_{ik}\right)\mathbf{C}_k^{-1}\mathbf{A}_k\right\} \tag{22}$$

- The gradient of $J_{\mathrm{DMC}}(\theta)$ with respect to the transformation matrix $\mathbf{A}_k$ is

$$\frac{\partial J_{\mathrm{DMC}}(\theta)}{\partial\mathbf{A}_k}=\frac{1}{n_k^2}\sum_{u,v}^{n_k}\frac{\exp(\beta L_{uv})}{1+\exp(\beta L_{uv})}l_{uv}\left\{\mathbf{P}_{uv}\mathbf{A}_k-\mathbf{P}_{uv}\mathbf{A}_k\mathbf{A}_k^T\mathbf{C}_k^{-1}\mathbf{A}_k-\mathbf{C}_k^{-1}\mathbf{A}_k\mathbf{A}_k^T\mathbf{P}_{uv}\mathbf{A}_k\right\} \tag{23}$$

Algorithm 1
The parameter optimization of DMR-MFA model.

Input: original dataset $\{\mathbf{x}_i\}_{i=1}^N$, the number of clusters K, subspace dimension m, threshold δ, and step size η;
Initialization: $\{(\mathbf{A}_k^{old}, \boldsymbol{\mu}_k^{old}, \boldsymbol{\Psi}_k^{old})_{k=1}^K, \boldsymbol{\pi}^{old}\}$;
Output: The optimal parameters $\{(\mathbf{A}_k^*, \boldsymbol{\mu}_k^*, \boldsymbol{\Psi}_k^*)_{k=1}^K, \boldsymbol{\pi}^*\}$;
Repeat:
$h_{ik} = E(\gamma_{ik}|X) = \frac{\pi_k^{old} N(\mathbf{x}_i; \boldsymbol{\mu}_k^{old}, \mathbf{A}_k^{old}(\mathbf{A}_k^{old})^T + \boldsymbol{\Psi}_k^{old})}{\sum_{k=1}^K \pi_k^{old} N(\mathbf{x}_i; \boldsymbol{\mu}_k^{old}, \mathbf{A}_k^{old}(\mathbf{A}_k^{old})^T + \boldsymbol{\Psi}_k^{old})}$;
Repeat:
$\nabla \mathbf{A}_k = \frac{\partial L_{\mathrm{MFA}}(\theta)}{\partial \mathbf{A}_k} - \lambda \frac{\partial J_{\mathrm{DMC}}(\theta)}{\partial \mathbf{A}_k}$; $\mathbf{A}_k^{new} = \mathbf{A}_k^{old} + \eta \nabla \mathbf{A}_k$;
$\nabla \boldsymbol{\Psi}_k = \frac{\partial L_{\mathrm{MFA}}(\theta)}{\partial \boldsymbol{\Psi}_k} - \lambda \frac{\partial J_{\mathrm{DMC}}(\theta)}{\partial \boldsymbol{\Psi}_k}$; $\boldsymbol{\Psi}_k^{new} = \boldsymbol{\Psi}_k^{old} + \eta \nabla \boldsymbol{\Psi}_k$;
$\nabla \boldsymbol{\mu}_k = \frac{\partial L_{\mathrm{MFA}}(\theta)}{\partial \boldsymbol{\mu}_k}$; $\boldsymbol{\mu}_k^{new} = \boldsymbol{\mu}_k^{old} + \eta \nabla \boldsymbol{\mu}_k$;
Until $\Delta\theta = \frac{\|\theta^{new} - \theta^{old}\|}{\|\theta^{old}\|} \le \delta, \forall \theta \in \{\mathbf{A}_k, \boldsymbol{\mu}_k, \boldsymbol{\Psi}_k\}_{k=1}^K$;
$\pi_k^{new} = \frac{1}{N} \sum_{i=1}^N h_{ik}$;
Until $\Delta Q = \frac{\|Q(\theta^{new}) - Q(\theta^{old})\|}{\|Q(\theta^{old})\|} \le \delta$;

- The gradient of $L_{\mathrm{MFA}}(\theta)$ with respect to the mean vector $\boldsymbol{\mu}_k$ is

$$\frac{\partial L_{\mathrm{MFA}}(\theta)}{\partial \boldsymbol{\mu}_k} = \sum_{i=1}^N h_{ik}(\mathbf{x}_i - \boldsymbol{\mu}_k)\mathbf{C}_k^{-1} \tag{24}$$

Based on above expressions, the optimization problem in (18) can be solved. The alternative iteration procedure is summarized in Algorithm 1, where δ denotes the convergence threshold, such as $\delta = 10^{-3}$ in our experiment, and η is the step size.

It should be pointed out that the inverse of $\mathbf{C}_k = (\mathbf{A}_k\mathbf{A}_k^T + \boldsymbol{\Psi}_k)$ $(k = 1, \cdots, K)$ need to be calculated in the model learning. However, the $\mathbf{A}_k\mathbf{A}_k^T$ will be a singular matrix when some rows of $\mathbf{A}_k$ are highly correlated. In this case, if diagonal covariance matrix $\boldsymbol{\Psi}_k$ contains zero (or extremely small) values at the diagonal elements corresponding to the rows with high similarity in $\mathbf{A}_k$, the determinant of $\mathbf{C}_k$ will be zero, thus making it impossible to calculate the inverse of $\mathbf{C}_k$ and hindering the implementation of our method. To deal with this problem, we can preset a small and nonzero threshold ρ (such as 10^{-5} in our experiment) and restricts the value of each diagonal element in $\boldsymbol{\Psi}_k$ larger than ρ during the iteratively updating process. With this approach, the calculation of $\mathbf{C}_k^{-1}$ has little effect on the applicability of our method.

3.3. The label prediction of a test sample

When the Algorithm 1 is convergent, the optimal parameters $\{\mathbf{A}_k^*, \boldsymbol{\mu}_k^*, \boldsymbol{\Psi}_k^*, \pi_k^*\}_{k=1}^K$ and the cluster index for each training sample can be obtained. Thereafter, the Bayesian classifier can be used to predict the class label of a test sample $\mathbf{x}^{test}$. In detail, the label prediction task consists of the following steps:

- Estimate the transformed low-dimensional latent features $\{\mathbf{y}_{uk}\}_{u=1}^{n_k}$ of the training data $\{\mathbf{x}_{uk}\}_{u=1}^{n_k}$ belonging to the kth cluster:

$$\mathbf{y}_{uk} = \left(\mathbf{A}_k^*\right)^T \left(\mathbf{C}_k^*\right)^{-1} \left(\mathbf{x}_{uk} - \boldsymbol{\mu}_k^*\right), k = 1, \ldots, K \tag{25}$$

- Compute the mean vector $\boldsymbol{\mu}_{yk}^c$ and covariance matrix $\boldsymbol{\Sigma}_{yk}^c$ based on the latent features with class label c in the kth cluster:

$$\boldsymbol{\mu}_{yk}^c = \frac{1}{n_k^c} \sum_{u=1}^{n_k^c} \mathbf{y}_{uk}^c, \boldsymbol{\Sigma}_{yk}^c = \frac{1}{n_k^c} \sum_{u=1}^{n_k^c} \left(\mathbf{y}_{uk}^c - \boldsymbol{\mu}_{yk}^c\right)\left(\mathbf{y}_{uk}^c - \boldsymbol{\mu}_{yk}^c\right)^T \tag{26}$$

where $\mathbf{y}_{uk}^c$ is the uth latent feature from class c in the kth cluster; $\boldsymbol{\mu}_{yk}^c$, $\boldsymbol{\Sigma}_{yk}^c$, and n_k^c are the mean vector, covariance matrix and number of the latent features with class label c in the kth cluster, respectively.

- Compute the cluster index z^{test} of a test sample $\mathbf{x}^{test}$ as

$$z^{test} = \underset{k \in \{1,2,\ldots,K\}}{\arg\max}\ \pi_k^* N\left(\mathbf{x}^{test}; \boldsymbol{\mu}_k^*, \mathbf{A}_k^*\mathbf{A}_k^{*T} + \boldsymbol{\Psi}_k^*\right) \tag{27}$$

- Estimate the latent feature $\mathbf{y}^{test}$ of $\mathbf{x}^{test}$ based on (13)

$$\mathbf{y}^* = \mathbf{A}_{z^{test}}^{*\,T}\left(\mathbf{A}_{z^{test}}^*\mathbf{A}_{z^{test}}^{*\,T} + \boldsymbol{\Psi}_{z^{test}}^*\right)^{-1}\left(\mathbf{x}^{test} - \boldsymbol{\mu}_{z^{test}}^{*\,T}\right) \tag{28}$$

- Compute the class label c^{test} of $\mathbf{x}^{test}$ via the Bayesian classifier

$$c^{test} = \arg\max_{c=1,2,\ldots,C}\left[-\frac{m}{2}\ln(2\pi) - \frac{1}{2}\ln\left|\boldsymbol{\Sigma}_{yk}^c\right| - \frac{1}{2}\left(\mathbf{y}^* - \boldsymbol{\mu}_{yk}^c\right)^T \boldsymbol{\Sigma}_{yk}^{c-1}\left(\mathbf{y}^* - \boldsymbol{\mu}_{yk}^c\right) + \ln(w_c)\right] \tag{29}$$

For an intuitive explanation, we give the classification flowchart based on the DMR-MFA (Fig. 4).

4. Experimental results

In this section, three types of datasets, including synthetic, benchmark and measured radar high-resolution range profile (HRRP) datasets, are utilized to assess the performance of the DMR-MFA, by comparing with the LDA, MMLDA, IDA, MIM, EIDA, LMNN, LLE, Isomap, KPCA, and MFA. In addition, for the MFA and our DMR-MFA, the number of clusters K should be preseted during the model

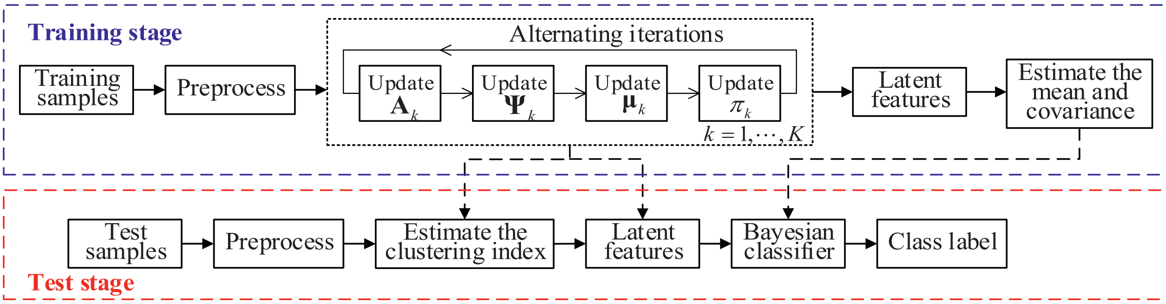


Fig. 4. Flowchart of the classification scheme based on the DMR-MFA. The training stage framed by the blue dotted line mainly performs the sample clustering and distribution estimation of latent feature in each cluster. The testing stage framed by the red dotted line first determines the cluster index of a test sample, and then its latent feature is learned, followed by the label prediction via the Bayesian classifier.

learning. In this paper, we adopt the cross validation method [28] to determine the K value. In detail, the training dataset is first divided into V subsets of equal size, and each subset is selected as the validation set in turn with the remaining subsets as the training sets. After obtaining V groups of training and validation sets, we pre-specify a candidate range of K, denoted as $\mathcal{M}$, and for each $K \in \mathcal{M}$, V models can be learned based on all groups of training set. Subsequently, the average classification performance of V models on the corresponding validation sets can be calculated for each $K \in \mathcal{M}$. Finally, the K_{opt} with the best classification performance is selected as an optimal estimation of K.

We comprehensively evaluate the models' classification performance via the measurement standards of classification accuracy, precision and recall, which are defined as

$$\text{accuracy} = \frac{n_{correct}}{N_{test}}, \ \text{precision} = \frac{1}{C}\sum_{c=1}^{C}\frac{n^c_{correct}}{N^c_{pred}}, \ \text{recall} = \frac{1}{C}\sum_{c=1}^{C}\frac{n^c_{correct}}{N^c}, \tag{30}$$

where $n_{correct}$, N_{test}, $n^c_{correct}$, N^c_{pred}, and N^c represent the numbers of correctly classified samples from all classes, test samples, correctly classified samples in class c, samples with predicted labels of c, and samples in the cth class, respectively. It should be noted that the experimental results on measured HRRP dataset are provided in the Supplementary Material due to the space limitation. Moreover, to obtain insights into the performance of the models in different classes, we also give the classification confusion matrices of all methods on each dataset, in Supplementary Material.

4.1. Synthetic data

4.1.1. Synthetic data with local linear separability

To evaluate the performance of the DMR-MFA on nonlinear separable dataset, we generate two classes of two-dimensional (2-D) data, each of which is of mixture Gaussian distribution with two modes. In detail, the means and covariances of two modes for the 1st class are set to

$$\boldsymbol{\mu}_1^1 = [-4, 4]^T, \ \boldsymbol{\Sigma}_1^1 = \begin{bmatrix} 0.7 & -0.5 \\ 0.5 & 0.7 \end{bmatrix}, \boldsymbol{\mu}_2^1 = [10, 4]^T, \ \boldsymbol{\Sigma}_2^1 = \begin{bmatrix} 0.7 & 0.5 \\ 0.5 & 0.7 \end{bmatrix}, \tag{31}$$

where $\boldsymbol{\mu}_1^1$ and $\boldsymbol{\Sigma}_1^1$ represent the mean and covariance of the 1st mode, respectively; $\boldsymbol{\mu}_2^1$ and $\boldsymbol{\Sigma}_2^1$ represent the mean and covariance of the 2nd mode. Similarly, the two modes of class 2 has the means $\boldsymbol{\mu}_1^2$, $\boldsymbol{\mu}_2^2$ and covariances $\boldsymbol{\Sigma}_1^2$, $\boldsymbol{\Sigma}_2^2$ as

$$\boldsymbol{\mu}_1^2 = [0, 4]^T, \ \boldsymbol{\Sigma}_1^2 = \begin{bmatrix} 0.7 & -0.5 \\ 0.5 & 0.7 \end{bmatrix}, \ \boldsymbol{\mu}_2^2 = [14, 4]^T, \boldsymbol{\Sigma}_2^1 = \begin{bmatrix} 0.7 & 0.5 \\ 0.5 & 0.7 \end{bmatrix}. \tag{32}$$

In the experiment, 200 samples are randomly sampled from each of two mixture Gaussian distributions in (31) and (32) with each mode containing 100 samples. Thereafter, we choose half of samples at regular intervals from each mode as the training data, and the other samples are used as the test data. To control the uncertainty caused by random sampling, the sampling work is repeated 10 times and the results are averaged over the 10 groups of samplings. Moreover, standard deviations are calculated to describe the uncertainty caused by multiple groups of samplings. In Fig. 5, we visualize the test data from two classes within a group of sampling, from which we see that samples are nonlinearly separable in the global space and can be divided into two locally linear separable regions.

Fig. 6 shows the clustering results of our method. We see that the globally nonlinear separable data are clustered into two linearly separable regions. Fig. 7 shows the 1-D latent features learned by differrent methods. Since above synthetic data is nonlinearly separable, the 1-D latent features learned via the LDA, MMLDA, IDA and LMNN are also linearly non-separable. Although LLE, Isomap and KPCA are nonlinear DR methods, these methods are not maximized the inter-class separability of features. Therefore, the 1-D latent features learned via these methods are also linearly non-separable. The 1-D features learned by our method are linearly separable due to its property of supervised nonlinear DR mechanism, which is benificial to the improvement on classification performance, as shown in Table 2.

4.1.2. Synthetic data with local nonlinear separability

Our DMR-MFA performs well on the globally nonlinear separable dataset, which is composed of several subsets with linear separability, such as the 100% classification accuracy on two classes of data generated by different GMMs in Fig. 5. To further evaluate the performance of our method on locally nonlinear separable data, we also generate two synthetic datasets, including concentric circles and local intersect circles, and then conduct the experiments on them.

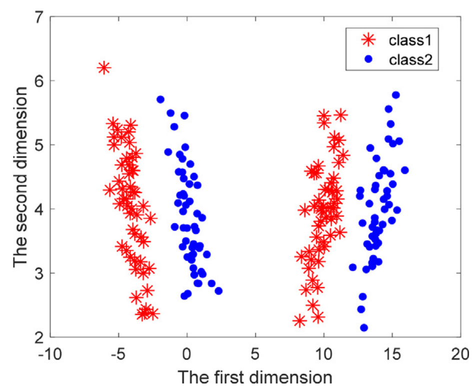


Fig. 5. The distribution of 2-D synthetic data from GMM. Here different colors represent different classes.

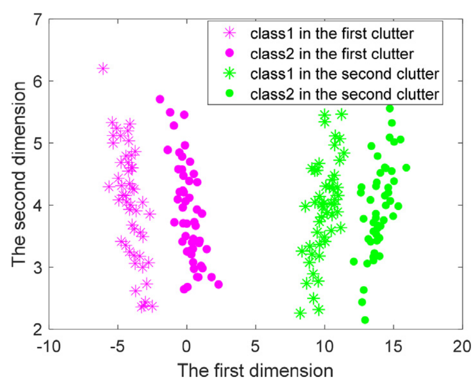


Fig. 6. The clustering result of our DMR-MFA model on synthetic data. The different colors represent different clusters.

Table 2
Comparison of classification performances on the synthetic data from mixture of Gaussian distributions via the different methods.

Methods	Accuracy (%)	Precision (%)	Recall (%)
DMR-MFA	***100±0.00***	***100±0.00***	***100±0.00***
LDA	50.45±0.02	50.60±0.03	50.45±0.02
MMLDA	49.87±0.02	50.20±0.03	49.87±0.02
IDA	50.75±0.02	50.10±0.04	50.75±0.02
MIM	50.75±0.04	50.00±0.02	50.75±0.04
EIDA	55.00±0.03	54.58±0.05	55.00±0.03
LMNN	49.58±0.04	50.58±0.03	49.58±0.04
LLE	70.50±0.04	80.37±0.03	70.50±0.04
Isomap	49.55±0.03	49.66±0.05	49.55±0.03
KPCA	90.50±0.02	92.02±0.03	90.50±0.02

Two classes of 2-D synthetic data are uniformly sampled from the ring and circle areas, respectively, with their centres being at the origin, the radiuses of internal and outer circles being 2 and 4 for the ring area, and the radius of circle area being 2. Moreover, each class contains 1000 samples, from which 500 samples are chosen at regular intervals and considered as the training data, with the remaining samples being the test data. Same as the experiment in Section 4.1.1, the sampling work is repeated 10 times. To give an intuitive illustration, we present the visualization of test data from two classes within one group of sampling in Fig. 8, from which we note that data from above two classes are locally nonlinear separable due to the circular structure of their classification surface. However, in principle, the samples from two classes in each local area are approximated to be linearly separable when the whole dataset is divided into enough subsets. Therefore, to obtain the best performance, the samples in each cluster should possesses the characteristic of approximately linear separation for our DMR-MFA model, in the case of optimal cluster number determined by the cross validation, and eventually, the proposed DMR-MFA can still achieve the good classification performance.

Table 3
Comparison of classification performances via different methods on the concentric circle dataset.

Methods	Accuracy (%)	Precision (%)	Recall (%)
DMR-MFA	***98.50±0.01***	***98.52±0.00***	***98.50±0.01***
LDA	79.60±0.04	79.99±0.03	79.60±0.04
MMLDA	77.20±0.02	77.60±0.04	77.20±0.02
IDA	77.80±0.03	78.36±0.02	77.80±0.03
MIM	88.60±0.05	89.14±0.03	88.60±0.05
EIDA	78.90±0.03	79.40±0.03	78.90±0.03
LMNN	70.40±0.03	70.44±0.02	70.40±0.03
LLE	80.20±0.02	80.88±0.01	80.20±0.02
Isomap	80.10±0.05	80.79±0.04	80.10±0.05
KPCA	79.40±0.03	79.99±0.03	79.40±0.03

To verify above analysis, Fig. 9 visualizes the clustering result of test samples under the condition of optimal cluster number. As shown in Fig. 9, the entire dataset is partitioned into 10 clusters, and the samples from different classes in each cluster (except the first and tenth cluster, each of which only contains one class of data) are nearly linear separable, especially for the third, sixth, seventh and eighth cluster. Due to this property of the proposed method on concentric circle dataset, our DMR-MFA model has the potential to achieve the better classification in each cluster and ultimately on the whole dataset.

Table 3 quantitatively compares the average classification performance of our model with other related methods over 10 groups of samplings, from which we see that the classification accuracy of DMR-MFA can be up to 98.50% and is far higher than those of related models. As a conclusion, our method still works for the locally nonlinear separable data like concentric circle.

Besides the locally nonlinear separable dataset where the boundaries of two classes are nonlinear and adjacent, another type of locally nonlinear separable data is the partial intersect dataset. For validating the performance of our method in this case, we generate two circles with their center coordinates being $(-2, 0)$, $(0, 0)$, and radiuses being 2, respectively. Then we sample 1000 data points from each circle and consider them belonging to different classes, with repeating 10 times sampling work. Similarly to the previous experiment, 500 samples for each class are selected at equal intervals as the training data and the remainder as the test data. In Fig. 10, we plot the distribution of test data within one sampling work, which clearly exhibits the nonlinear separability in the intersect area of two classes of data.

We learn our DMR-MFA model using above training samples and perform the clustering operation via the learned DMR-MFA on test data in Fig. 10. The clustering result under optimal cluster number is present in Fig. 11. As shown in Fig. 11, the dataset is totally divided into four clusters, of which the first and second clusters contain different classes of data. Since the samples in these two clusters are mixed together, the features extracted by our DMR-MFA model are of nonlinear separation due to the linear DR method via a FA model in each cluster. Therefore, the proposed method performs bad in the first and second clusters, and the classification performance on the whole dataset will thus worse than that on the concentric circle dataset, such as the average classification accuracy of 92.90% shown in Table 4.

Actually, neither our DMR-MFA model nor other related methods can extract the linearly separable features from the intersect area due to the property of data distribution. However, superior to the linear DR methods, our model has the ability to describe ob-

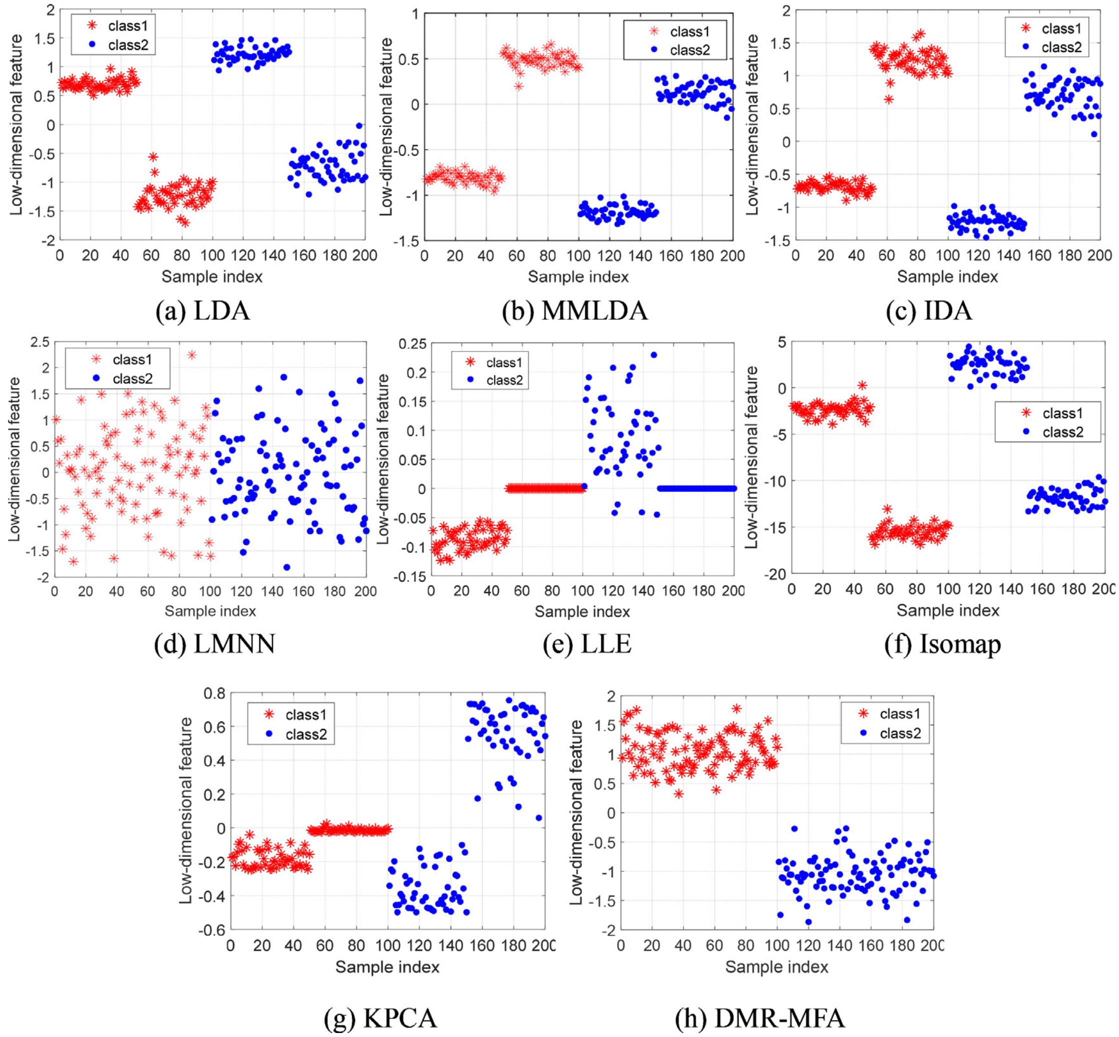


Fig. 7. The 1-D latent features of synthetic data in Fig. 5, learned by the LDA, MMLDA, IDA, LMNN, LLE, Isomap, KPCA and DMR-MFA. For each subfigure, the different colors represent different classes.

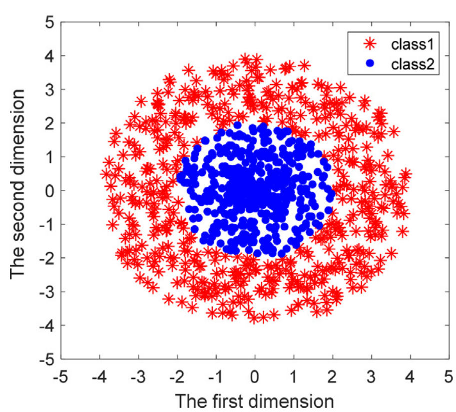


Fig. 8. Distribution of the concentric circle data, with different colors representing different classes.

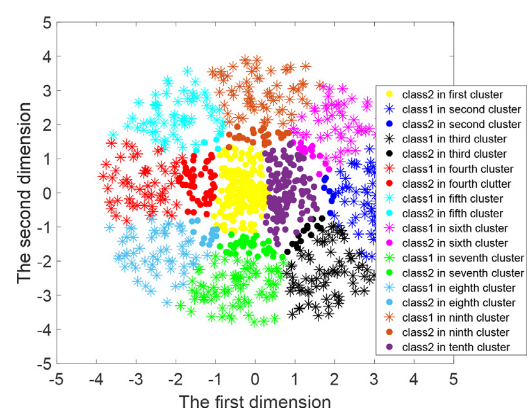


Fig. 9. Clustering result of data in Fig. 8, via the DMR-MFA. Different colors represent different clusters.

servations' statistical characteristics. Moreover, compared with the unsupervised nonlinear DR methods, the DMR-MFA introduces the supervised constraint and promotes the inter-class feature differences. Our method can obtain the better performance on the disjoint data areas than other methods, ultimately the higher classification accuracy on the whole intersect circle dataset, as shown in Table 4.

4.2. Benchmark datasets

We further evaluate the effectiveness of our DMR-MFA model on five benchmark datasets, including Splice, Flare_solar, Waveform, X8D5K and Satimage, of which the data introductions and 2-D visualizations via the t-distribution Stochastic Neighbor Embedding (t-SNE) [29] are given in Table 5 and Fig. 12, respectively.

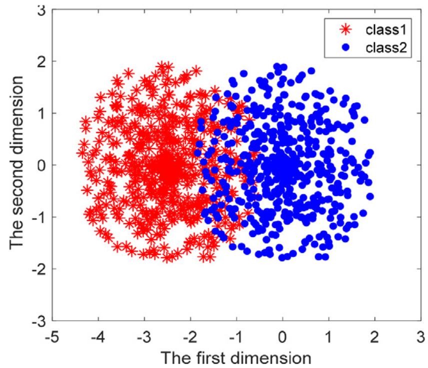


Fig. 10. Distribution of the local intersect circle data, with different colors representing different classes.

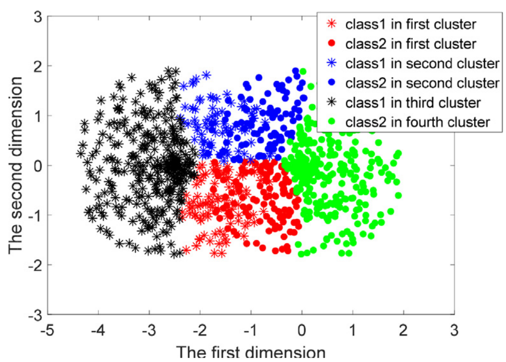


Fig. 11. Clustering result of test data in Fig. 10, via our DMR-MFA model. Different colors represent different clusters.

Table 4
Comparison of classification performances via different methods on the local intersect circle dataset.

Methods	Accuracy (%)	Precision (%)	Recall (%)
DMR-MFA	***92.90±0.01***	***92.90±0.01***	***92.90±0.01***
LDA	88.30±0.05	89.00±0.04	88.30±0.05
MMLDA	88.80±0.04	88.80±0.05	88.80±0.04
IDA	88.80±0.05	88.90±0.04	88.80±0.05
MIM	88.60±0.06	88.70±0.04	88.60±0.06
EIDA	88.10±0.07	88.50±0.06	88.10±0.07
LMNN	86.10±0.04	86.20±0.03	86.10±0.04
LLE	88.60±0.03	89.00±0.02	88.60±0.03
Isomap	88.90±0.04	89.00±0.04	88.90±0.04
KPCA	89.65±0.03	89.70±0.02	89.65±0.03

Table 5
Data introduction of five benchmark datasets used in the experiments.

Name	The number of classes	Data dimension	The number of training samples	The number of test samples
Splice	2	60	1000	2175
Flare_solar	2	9	666	400
Waveform	3	21	2500	2500
X8D5K	5	8	500	500
Satimage	6	36	4435	2000

The classification performances of our DMR-MFA and other related methods on five benchmark datasets are compared in Fig. 13. From Fig. 13 we see that that our DMR-MFA obtains the best performances, including classification accuracy, precision and recall, on five benchmark datasets. Compared with the supervised linear DR methods, i.e., LDA, MMLDA, IDA, MIM, EIDA and LMNN, our model can also effectively deal with the problem of DR from nonlinear separable data, thus offering the potential to perform better on a variety of datasets. Compared with other four unsupervised nonlinear DR methods, i.e., LLE, Isomap, KPCA, and MFA, our DMR-MFA model introduces the supervised DMC, which further enhances the separability of the transformed low-dimensional data from different classes, ultimately leading to the higher classifica-

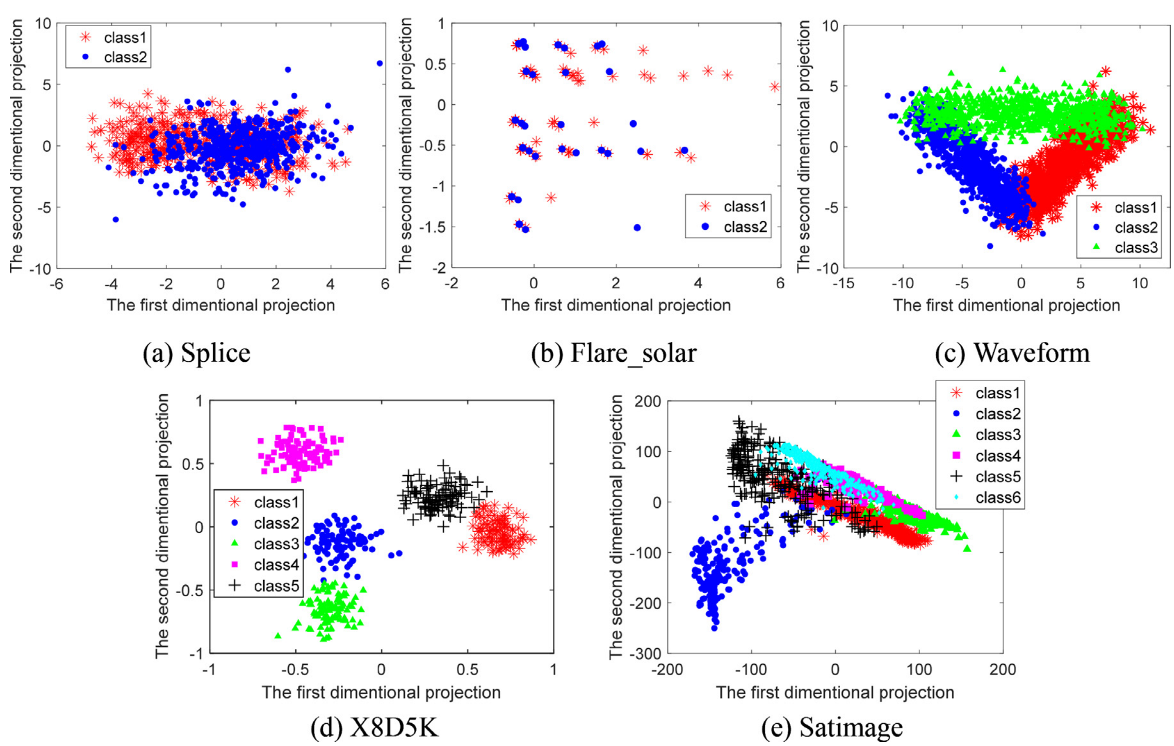


Fig. 12. 2-D visualizations of test data on five benchmark datasets via the t-SNE method, in which different colors represent different classes.

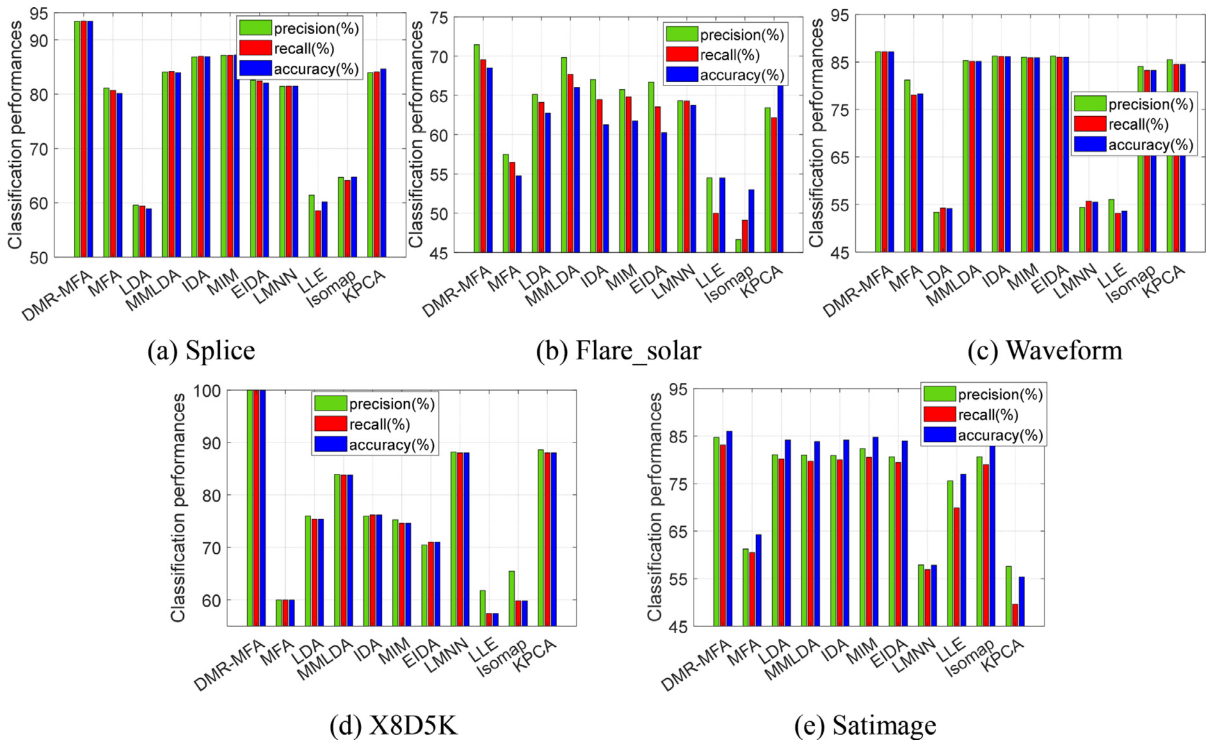


Fig. 13. The comparisons of classification accuracy, precision, and recall on five benchmark datasets, via the DMR-MFA model and other related methods.

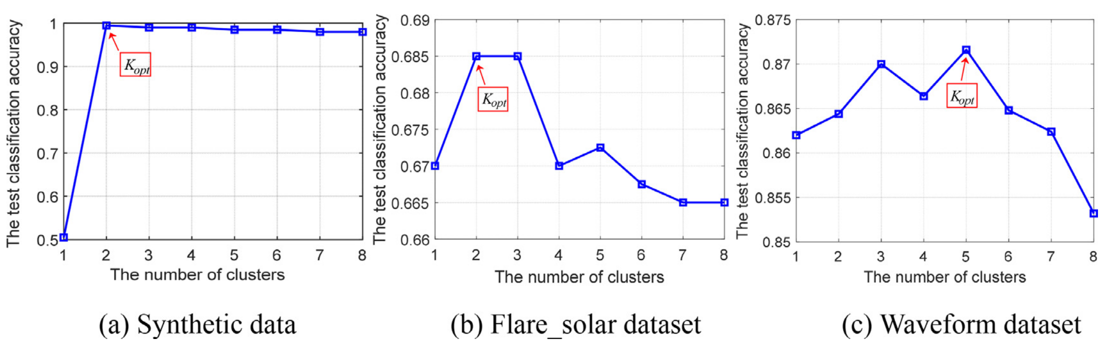


Fig. 14. The test classification accuracies of the DMR-MFA model on three datasets, under the condition of different cluster numbers. The K_{opt} in the red box corresponds to the optimal number of clusters determined by the cross validation method.

tion precision. Therefore, the proposed DMR-MFA model is more suitable for the DR of high-dimensional data.

From Fig. 13 we also find that the our model can make the completely correct predictions for test data from X5D8K. This is because it is approximately linear separable in local area as shown in Fig. 12(d), which is coincident with the scene that our model is most applicable to as demonstrated in Section 4.1. For Splice, Waveform and Satimage datasets, the data from different classes are local intersect, and consequently, their classification performance via the DMR-MFA is weaker than the X5D8K. However, for Flare_solar, there is a larger intersect area for the data from different classes than the other four datasets, which thus leads to the worst classification accuracy of our model on Flare_solar.

4.3. Analysis of hyperparameter

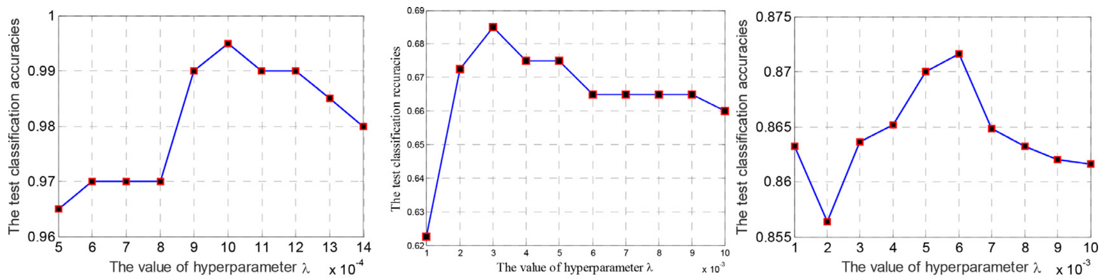
4.3.1. The influence of cluster number

The number of clusters K in our DMR-MFA model has a great effect on the classification performance. The K value that is too large will lead to the higher model complexity, thus easily causing the overfitting and weakening the model generalization; on the contrary, the too small value of K is not conducive to obtain the more separable data subset and eventually the features with greater inter-class difference. As demonstrated at the beginning of this section, we determine the optimal cluster number K via the cross validation method. To demonstrate the effectiveness of cross validation on the determination of cluster number K, we take synthetic data in Fig. 5, Flare_solar and Waveform datasets as examples and present the classification accuracies of our DMR-MFA model on test samples of above datasets, under the condition of different K values in Fig. 14, where the K_{opt} represents the optimal number of clusters determined by the cross validation.

As we can see from Fig. 14, the test classification accuracy varies with different K values, and an inappropriate setting of K may lead to the bad classification performance. As an alternative way, the cross validation approach is capable of determining a better K value, and thus can be used for the estimation of cluster number in our method.

4.3.2. The influence of weight

The hyperparameter λ in our DMR-MFA controls the proportion of the distance metric constraint in the MFA model, and the per-



(a) Synthetic data (b) Flare_solar (c) Waveform

Fig. 15. The variations of classification accuracies of our method with hyperparameter λ on two classes of mixture Gaussian synthetic data, Flare_solar and Waveform datasets, respectively.

formance of our method will be poor if λ is set too large (in this case, our model degenerates into the Mahalanobis distance metric learning and cannot extract the better features from nonlinear separable data) or too small (in this case, the distance metric constraint loss function has little effect on the parameter optimization, thus weakening the distinctions of features from different classes). To more intuitively illustrate the effect of λ on the classification performance, we take two classes of mixture Gaussian synthetic data in Fig. 5, Flare_solar and Waveform datasets as examples and present their variation curves of test classification accuracies with the value of λ in Fig. 15.

We see from Fig. 15 that the value of λ significantly affects the classification accuracy of our method, and either too high or too low λ is unsuitable. In addition, the classification accuracies of the proposed model are the best when λ is equal to 0.001, 0.003 and 0.006 for mixture Gaussian synthetic data, Flare_solar and Waveform datasets, respectively, which demonstrates that λ value should be adjusted for different datasets.

5. Conclusion

Aiming for the extraction of low-dimensional features with more inter-class separability, we propose a DMR-MFA model. The DMR-MFA partitions the whole dataset into several subsets and individually performs linear DR with a FA model on each subset. Due to the collection of multiple FA models, our model is a nonlinear DR method and more suitable for the DR under the condition of complex data distribution (such as the globally nonlinear separable but locally linear separable data) than the linear DR methods. Meanwhile, the DMR-MFA promotes the feature learning in each subset toward the direction of more separable by introducing the distance metric constraint, which further ensures the separability of features from different classes, and thus outperforms the unsupervised nonlinear DR methods. In summary, thanks to the stronger capability of MFA in data description and the auxiliary function of distance metric constraint in more separable feature learning, our DMR-MFA has much superiority over the traditional linear DR and unsupervised nonlinear DR methods. Along this line, a class of approaches combining the distance metric learning and unsupervised nonlinear model can be developed to tackle the DR problem. For example, we can replace the MFA used in our model with KPCA, autoencoder (AE) [7] , variational autoencoder (VAE) [30] , and so on.

The main weakness of our method is the determination of the cluster number K, which has significant effect on the feature separability as illustrated in Section 4.3.2 and is determined based on the cross validation. Nevertheless, the cross validation has the limitation in efficiency and requires a large demand for the storage space. Therefore, besides the study on the combination of other nonlinear DR methods and distance metric learning, we will also explore the solution to the automatic determination of cluster number in our future work, such as imposing the Dirichlet process mixture (DPM) prior [31,32] on the cluster indicator γ_{ik} by referencing the nonparametric Bayesian models.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.patcog.2021.108156.

References

[1] X.B. Shen, Y.H. Yuan, F.M. Shen, Y. Xu, Q.S. Sun, A novel multi-view dimensionality reduction and recognition framework with applications to face recognition, J. Vis. Commun. Image Represent. 53 (2018) 161–170, doi:10.1016/j.jvcir.2018.03.004.
[2] Z.Z. Feng, M. Yang, L. Zhang, Y. Liu, D. Zhang, Joint discriminative dimensionality reduction and dictionary learning for face recognition, Pattern Recognit. 46 (8) (2013) 2134–2143, doi:10.1016/j.patcog.2013.01.016.
[3] F.L. Luo, L.P. Zhang, B. Du, L.F. Zhang, Dimensionality reduction with enhanced hybrid-graph discriminant learning for hyperspectral image classification, IEEE Trans. Geosci. Remote Sens. 58 (8) (2020) 5336–5353, doi:10.1109/TGRS.2020.2963848.
[4] D.F. Gillies, R. Liu, Overfitting in linear feature extraction for classification of high-dimensional image data, Pattern Recognit. 53 (2016) 73–86, doi:10.1016/j.patcog.2015.11.015.
[5] J. Sanchez, F. Perronnin, High-dimensional signature compression for large-scale image classification, IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2011 https://doi.org/2011. 10.1109/CVPR.2011.5995504.
[6] C. Du, B. Chen, B. Xu, D.D. Guo, H.W. Liu, Factorized discriminative conditional variational auto-encoder for radar HRRP target recognition, Signal Process. 158 (2019) 176–189, doi:10.1016/j.sigpro.2019.01.006.
[7] B. Feng, B. Chen, H.W. Liu, Radar HRRP target recognition with deep networks, Pattern Recognit. 61 (2017) 379–393, doi:10.1016/j.patcog.2016.08.012.
[8] A. Akusok, E. Eirola, Comparison of classification methods for very high-dimensional data in sparse random projection representation, in: Proceedings of Extreme Learning Machine (ELM), 2018, doi:10.1007/978-3-030-23307-5_3.
[9] Q.Q. Wamg, Q.X. Gao, X.B. Gao, F.P. Nie, $l_{2,p}$-norm based PCA for image recognition, IEEE Trans. Image Process. 27 (3) (2018) 1336–1346, doi:10.1109/TIP.2017.2777184.
[10] B. Ghaddar, J. Naoum-Sawaya, High dimensional data classification and feature selection using support vector machines, Eur. J. Oper. Res. 265 (2018) 993–1004, doi:10.1016/j.ejor.2017.08.040.
[11] A. Subasi, M.I. Gursory, EEG signal classification using PCA, ICA, LDA and support vector machines, Expert Syst. Appl. 37 (12) (2010) 8659–8666, doi:10.1016/j.eswa.2010.06.065.
[12] J. Xu, J. Yang, Z.H. Gu, N. Zhang, Median-mean line based discriminant analysis, Neurocomputing 123 (2014) 233–246, doi:10.1016/j.neucom.2013.07.012.

[13] Z. Nenadic, Information discriminant analysis: feature extraction with an information-theoretic objective, IEEE Trans. Pattern Anal. Mach. Intell. 29 (8) (2007) 1394–1407, doi:10.1109/TPAMI.2007.1156.
[14] K.Q. Weinberger, L.K. Saul, Distance metric learning for large margin nearest neighbor classification, J. Mach. Learn. Res. 10 (2009) 207–244, doi:10.1007/s10845-008-0108-2.
[15] C.C. Chang, A boosting approach for supervised Mahalanobis distance metric learning, Pattern Recognit 45 (2) (2012) 844–862, doi:10.1016/j.patcog.2011.07.026.
[16] M.E. Hellman, J. Raviv, Probability of error, equivocation and the chernoff bound, IEEE Trans. on Inf. Theory 16 (1970) 368–372, doi:10.1109/TIT.1970.1054466.
[17] M. Chen, W. Carson, M. Rodrigues, R. Calderbank, L. Carin, Communications inspired linear discriminant analysis, in: Proceedings of the 29th International Conference on Machine Learning, Edinburgh, Scotland, UK, 2012, pp. 919–926.
[18] L.L. Li, L. Du, W. Zhang, H. He, P.H. Wang, Enhancing information discriminant analysis, Pattern Recognit. 60 (C) (2016) 554–570, doi:10.1016/j.patcog.2016.06.004.
[19] Z. Zhang, T.W.S. Chow, M.B. Zhao, M-Isomap: orthogonal constrained marginal isomap for nonlinear dimensionality reduction, IEEE Trans. Cybern. 43 (1) (2013) 180–191, doi:10.1109/TSMCB.2012.2202901.
[20] H. Rajaguru, S.K. Prabhakar, Performance analysis of local linear embedding (LLE) and Hessian LLE with hybrid ABC-PSO for epilepsy classification from EEG signals, International Conference on Inventive Research in Computing Applications (ICIRCA), 2018.
[21] Y.C. Motai, Kernel association for classification and prediction: a survey, IEEE Trans. Neural Netw. Learn. Syst. 26 (2) (2015) 208–223, doi:10.1109/TNNLS.2014.2333664.
[22] F. Wang, S. Zhang, Y.X. Yin, Log likelihood monitoring for multimode processing using variational Bayesian mixture factor analysis model, IEEE Access 7 (2019) 2169–3536, doi:10.1109/ACCESS.2019.2925884.
[23] E.W. Dijkstra, A note on two problems in connexion with graphs, Numerische Mathematik 1 (1959) 269–271, doi:10.1007/BF01386390.
[24] R.W. Floyd, Algorithm 97: shortest path, Commun. ACM 5 (6) (1962) 345, doi:10.1145/367766.368168.
[25] X.F. Zhang, B. Chen, H.W. Liu, L. Zuo, B. Feng, Infinite max-margin factor analysis via data augmentation, Pattern Recognit. 52 (2016) 17–32, doi:10.1016/j.patcog.2015.10.020.
[26] A.S. Rahmathulah, R. Selvan, L. Svensson, A batch algorithm for estimating trajectories of point targets using expectation maximization, IEEE Trans. Signal Process. 64 (18) (2016) 4792–4804, doi:10.1109/tsp.2016.2572048.
[27] J. Chen, L. Du, H. He, Y.C. Guo, Convolutional factor analysis model with application to radar automatic target recognition, Pattern Recognit. 87 (2019) 140–156, doi:10.1016/j.patcog.2018.10.014.
[28] T.T. Wong, N.Y. Yang, Dependency analysis of accuracy estimates in k-fold cross validation, IEEE Trans. Knowl. Data Eng. 29 (11) (2020) 2417–2427 http://doi.org/10.1109/TKDE.2017.2740926.
[29] L. van der Maaten, G. Hinton, Visualizing data using t-SNE, J. Mach. Learn. Res. 9 (2008) 2579–2605 http://doi.org/10.1007/s10846-008-9235-4.
[30] D.P. Kingma, M. Welling, Auto-encoding variational Bayes, in: Proceedings of the International Conference on Learning Representations (ICLR), 2014.
[31] E. Abbasnejad, A. Dick, V.D.A. Hengel, Infinite variational autoencoder for semi-supervised learning, in: Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, doi:10.1109/CVPR.2017.90.
[32] W. Zhang, L. Du, L.L. Li, X.F. Zhang, H.W. Liu, Infinite Bayesian one-class support vector machine based on Dirichlet process mixture clustering, Pattern Recognit. 78 (2018) 56–78, doi:10.1016/j.patcog.2018.01.006.

Jian Chen received the B.Eng. degree in communication engineering from the Qingdao University of Technology, Qingdao, China, in 2014, and the Ph.D. degree in electronic engineering from Xidian University, Xi'an, China, in 2020. His research interests include radar signal processing, radar automatic target recognition, and pattern recognition.

Leiyao Liao received the B.Eng. degree in electrical engineering from Xidian University, Xi'an, China, in 2017, where she is currently pursuing the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing. Her research interests include radar automatic target recognition and pattern recognition.

Wei Zhang received the B.Eng. and Ph.D. degrees in electronic and information engineering from Xidian University, Shanxi, Xi'an, China, in 2014 and 2020. His research interests include radar signal processing and pattern recognition.

Lan Du received the B.S., M.S. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in Jul. 2001, Mar. 2004 and Jun. 2007 respectively. Her doctoral dissertation was granted National Excellent Doctoral Dissertation of PR China in 2009. From Sep. 2007 to Sep. 2009, she did research work as a Postdoctoral Research Associate in the Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708 USA. She is currently a Professor at National Laboratory of Radar Signal Processing, Xidian University. Her research work is supported by NSFC for Excellent Young Scholars. Her main research interests are in the fields of statistical signal processing and machine learning with application to radar target recognition.